The Medallion Architecture: Data Layer Comparison

Data Engineering Concepts

Overview of the Medallion Architecture

The Medallion Architecture is a data design pattern used to logically organize data in a lakehouse or data warehouse. Its goal is to incrementally and progressively improve the structure and quality of data as it flows through multiple layers: **Bronze** (Raw), **Silver** (Cleansed), and **Gold** (Curated).

1. Bronze Layer (Raw Data)

The Bronze layer is the landing zone for all data extracted from external source systems. The primary focus here is quick, resilient data ingestion.

- **Data State:** Raw, unparsed, and untransformed.
- **Structure:** Retains the exact schema and format of the source system (e.g., JSON, CSV, API payloads).
- **Purpose:** Acts as a historical archive and a safety net. If a transformation pipeline breaks downstream, you can always replay the data from the Bronze layer.
- **Operations:** Append-only. Data is rarely updated or deleted here.

2. Silver Layer (Cleansed & Conformed Data)

The Silver layer takes the raw data from the Bronze layer and applies quality checks, filtering, and standardization. It serves as the enterprise's central "Source of Truth."

- **Data State:** Filtered, cleaned, and normalized.
- **Structure:** Data is parsed into structured tables. Column names are standardized, and data types are enforced.
- **Purpose:** To provide a consistent, unified view of the core entities (e.g., Customers, Products, Transactions) across the organization.
- **Operations:** Deduplication, handling missing/null values, standardizing dates, and resolving schema drift.

3. Gold Layer (Curated Business-Level Data)

The Gold layer is highly refined and optimized specifically for business intelligence (BI), analytics, and machine learning models.

- **Data State:** Aggregated, enriched, and business-ready.

- **Structure:** Typically modeled using a Star Schema (Fact and Dimension tables) or highly denormalized wide tables for performance.
- **Purpose:** Designed to answer specific business questions quickly. This is what end-users, dashboards, and reporting tools connect to.
- **Operations:** Complex table joins, business logic application, metric aggregations (e.g., monthly sales totals), and surrogate key generation.

Summary Comparison Table

Feature	Bronze (Raw)	Silver (Cleansed)	Gold (Curated)
Primary Goal	Fast ingestion & historical storage	Data quality, standardization, & integration	Business analytics & fast querying
Data Quality	Low (Raw, messy)	Medium to High (Cleaned, deduplicated)	Very High (Business rules applied)
Data Structure	Source-aligned (JSON, CSV, Raw SQL)	Normalized, structured tables	Star schema (Facts/Dims), denormalized
Target Audience	Data Engineers	Data Engineers, Data Scientists	Data Analysts, BI Tools, Business Users
Operations	Append-only (Insert)	Merge, Update, Cleanse, Filter	Aggregate, Join, Calculate Metrics